

Notes on Reproducing the Single-Bearing EqF Example

Reproducibility note

May 1, 2026

1 Purpose

This note summarizes a reproduction study of the single-bearing estimation example from the Equivariant Filter (EqF) paper by van Goor, Hamel, and Mahony. The goal is not to dismiss the value of equivariant filtering. Rather, it is to understand exactly what is being compared, what assumptions are required for the theory to apply, and what happens when those assumptions are weakened.

The example is especially useful pedagogically because it is simple enough to analyze, but subtle enough to expose several common pitfalls in simulation comparisons.

This is an independent reproduction exercise, not a definitive audit of the authors' code. Some discrepancies below may be due to our implementation choices, unpublished tuning values, or convention differences. The appropriate conclusion is that the example deserves careful checking and that the published description is difficult to reproduce exactly.

2 The Bearing Problem Is Not the Clean IEKF Case

The strongest invariant-EKF intuition applies when the state itself is a Lie group and the dynamics are group-affine. In that setting, an invariant error can sometimes be mapped through the logarithm to a vector space where the deterministic error dynamics are exactly linear.

The single-bearing problem is different. The state is

$$\eta \in S^2, \quad \dot{\eta} = -\Omega^\times \eta, \quad y = c_m \eta.$$

The sphere is a homogeneous space of $SO(3)$, but it is not itself a Lie group. The action of $SO(3)$ on S^2 is transitive but not free: many rotations produce the same bearing. Thus there is no global bijection between $\mathfrak{so}(3)$ and S^2 .

The EqF construction still provides a useful equivariant error

$$e = \phi(\hat{R}^{-1}, \eta) = \hat{R}\eta,$$

and with the paper's normal coordinates the preobserver dynamics are simple. However, the measurement update remains nonlinear in the chosen error coordinates. This is where much of the comparison lives.

3 Output Linearization: What EqF* Changes

At the origin, write the normal-coordinate error as

$$\epsilon = \begin{bmatrix} a \\ b \end{bmatrix}, \quad \rho = \|\epsilon\|.$$

The inverse normal chart is

$$\eta(\epsilon) = \begin{bmatrix} \cos \rho \\ -b \frac{\sin \rho}{\rho} \\ a \frac{\sin \rho}{\rho} \end{bmatrix}.$$

The exact residual is

$$r(\epsilon) = \eta(\epsilon) - e_1.$$

The standard EqF output matrix gives

$$C\epsilon = \begin{bmatrix} 0 \\ -b \\ a \end{bmatrix}, \quad r(\epsilon) - C\epsilon = O(\rho^2).$$

The EqF* output matrix cancels the second-order term:

$$r(\epsilon) - C^*\epsilon = O(\rho^3).$$

This is a real local advantage. However, EqF* is also a measurement-dependent output approximation. It is not simply the same Kalman filter applied in better coordinates. It modifies the output model in a way that is not applied symmetrically to the EKF baseline in the paper.

This matters. When comparing filters, readers should ask whether each method has been given a comparable “enhancement budget.” If one method receives a measurement-dependent correction to its output Jacobian, a fair comparison should consider analogous corrections for the other methods.

4 A Sign Inconsistency in the EKF Derivation

The paper defines the embedded EKF error as

$$\tilde{\eta} := \eta - \hat{\eta}.$$

For the norm constraint

$$g(\eta) = \|\eta\|^2 = 1,$$

the first-order expansion is

$$g(\eta) - g(\hat{\eta}) = 2\hat{\eta}^\top (\eta - \hat{\eta}) + O(\|\tilde{\eta}\|^2).$$

Thus the constraint Jacobian should have the positive sign

$$G = 2\hat{\eta}^\top.$$

The paper text writes this term with the opposite sign. The published Figure 3 EKF heatmap also appears to match the opposite residual sign from the one stated in the text. This does not prove that the authors’ simulation code used the wrong sign; it may be only a manuscript or plotting-convention issue. Nevertheless, it is a concrete inconsistency worth checking in any reproduction.

Figure 1 compares the two EKF output-linearization sign conventions.

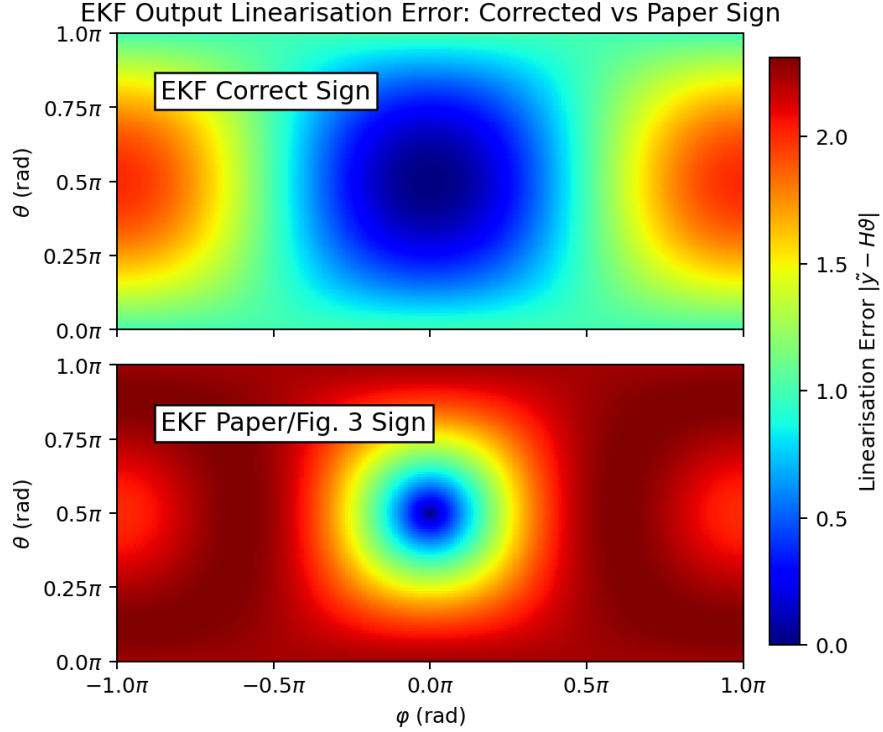


Figure 1: EKF output-linearization error for the mathematically corrected sign and the opposite sign convention that appears to match the paper’s Figure 3.

5 Reproduction Findings

The paper states the main noise levels and initial distribution, but it does not publish all numerical tuning values used for the Riccati terms M_ϵ , N_ϵ , or the EKF virtual constraint covariance. These values matter. With one reasonable reconstruction of the experiment, using 500 Monte Carlo trials, the corrected EKF is much closer to EqF* than the paper’s qualitative conclusion suggests. These numbers should be read as evidence that the comparison is sensitive to implementation details, not as a claim that our reconstruction exactly matches the authors’ experiment.

Figure 2 shows the corresponding Monte Carlo comparison. The flipped-sign EKF is included as a diagnostic, not as a recommended algorithm.

6 What the Adaptive Tests Show

We tested a measurement-dependent secant matrix C_{sec} that fits the current residual direction. In the noiseless deterministic approximation, this can remove residual truncation error along the measured displacement. But pure secant is rank one and very sensitive to measurement noise near convergence.

A noise-aware blend

$$C_{\text{ad}} = (1 - \alpha)C + \alpha C_{\text{sec}}, \quad \alpha = \frac{\rho^2}{\rho^2 + k\sigma_y^2},$$

performed better than pure secant. This rule gives secant information more weight when the residual is large relative to measurement noise, and less weight near convergence.

Method	Final median angle error (deg)	Final median Lyapunov value
EqF	0.2713	9.52×10^{-4}
EqF*	0.1956	4.94×10^{-4}
Adaptive EqF secant, $k = 8$	0.1903	4.27×10^{-4}
Corrected embedded EKF	0.2081	9.75×10^{-4}
Adaptive embedded EKF, $k = 8$	0.1922	5.90×10^{-4}
EKF with flipped constraint sign	0.3582	3.42

Table 1: Representative 500-trial reproduction results. The adaptive methods use the same noise-aware secant blending idea for the EqF and EKF output matrices.

The important lesson is that this idea is not unique to the EqF. Applying an analogous adaptive secant correction to the embedded EKF also improved the EKF substantially. Therefore, part of the observed improvement comes from the output approximation, not only from the equivariant lift.

7 Conclusions

1. Symmetry and equivariance can produce better error coordinates, but the strongest IEKF-style “exact linear error” story does not fully apply to a homogeneous space such as S^2 .
2. EqF* is a useful output approximation, but it is also an additional design choice. It should not be treated as a purely automatic consequence of using a better coordinate system.
3. The paper appears to contain at least one sign inconsistency in the EKF derivation, and the Figure 3 EKF panel appears to use the opposite residual sign from the text. This should be checked against the authors’ actual simulation code before drawing a stronger conclusion.
4. Missing simulation settings make exact reproduction difficult. In nonlinear filtering, tuning values and discretization choices can materially affect the conclusion.
5. A corrected EKF baseline is much more competitive than the paper’s qualitative narrative suggests in our reconstruction. This does not invalidate EqF, but it does suggest that the strongest performance claims for this particular example should be interpreted with care.
6. The broader research lesson is constructive: when a method wins, ask whether the gain comes from geometry, from an output approximation, from tuning, or from an unfair baseline. Good papers make this separation easy to audit.

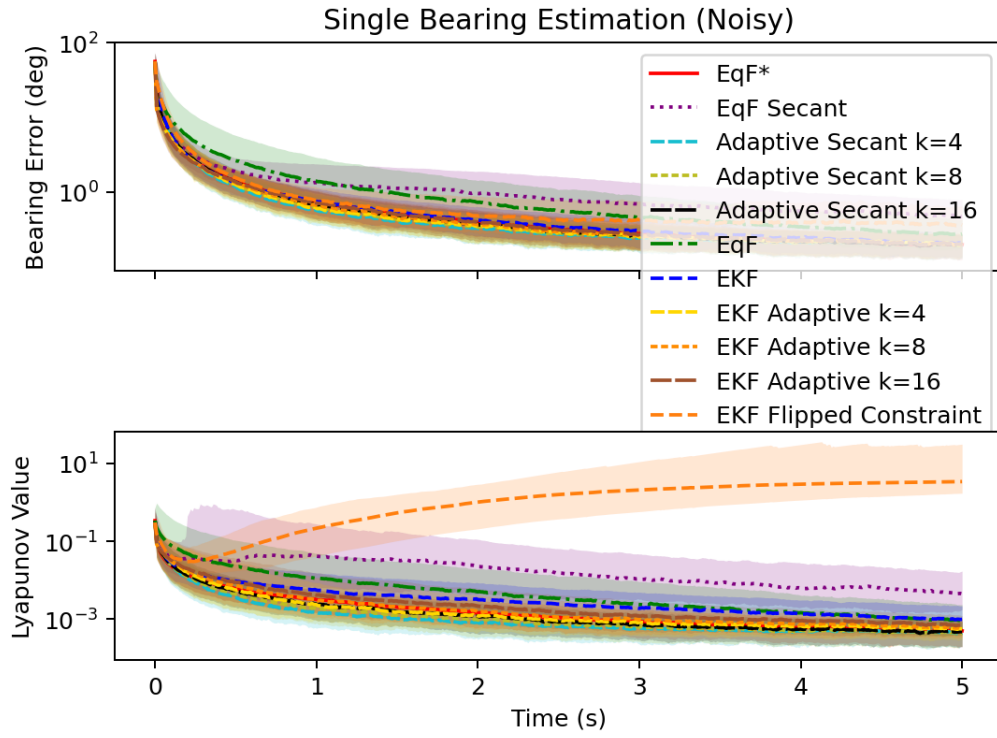


Figure 2: Monte Carlo comparison including EqF, EqF*, corrected EKF, flipped-sign EKF, and adaptive output-Jacobian variants.